

Gate E6 report — run_v1.21_r1 (Paper IV v1.21)

Verdict: **PASS with findings**. Evidence: 20_EVIDENCE/E6/.

E6 — Bilingual artifacts revalidation (v1.21)

Coverage.

- Identities: all six target files (EN/ES × MD/TeX/PDF) match `E0_INITIAL.json`. Page counts are EN 71 and ES 72 (pymupdf).
- Automated parity: done by the delegated sub-agent (same model) before two rate-limit interruptions. Its outputs are in `results/`: equation/section reference parity, key-number counts across the four sources, a prime check, headings set as code, a "holgura" scan, line alignment, and PDF text and font checks.
- Page inspection: after the second interruption the lead auditor rendered both PDFs again at 90 dpi as 2-up sheets (`scripts/make_pairs.py, pairs/`) and viewed every sheet itself: **143/143 pages** (71 EN + 72 ES), logged in `PAGE_INSPECTION_LOG.csv`. Zooms in `results/` (binder, figures) were used where needed. Because of the 90 dpi 2-up render, fine glyph defects on dense pages may have been missed.

Parity.

- Key numbers of the new passages (338, 1388, 118, 238, 2208, 59485, $a_0 + 31$, 191424, 12748, 12749, 392, $\rho/40$, $\rho/200$, 10^{10} , $n/9$, $4n/10^4$, 2^{-59485}) occur the same number of times in EN.md, ES.md, EN.tex and ES.tex.
- No "§3.2" remains. C' is a typographic prime in MD; the TeX uses C' in math (renders as a prime in both PDFs).
- The Figure 2 reference (EN p.15 → figure on p.16; ES p.16) and the new Figure 3 citation in §6.6 (EN p.25 → p.26; ES p.26) resolve correctly.
- The page-1 status paragraph is accurate in both editions.

Resolution of earlier findings

ID	Status	Evidence
X-04 "holgura" for the margin	RESOLVED	ES uses "margen" for $\Delta(G)$ everywhere (incl. §3 title "régimen con margen"); "holgura" survives only as "holgura del selector" (l.159) and for the slack nibble (Lemma 3.2 title, l.1461), which is the selector-slack notion.
X-12 Lean identifiers in roman / unmarked breaks	PARTIAL	Breaks now carry the continuation arrow throughout (e.g. EN pp.17, 22, 31–33, 39, 64; ES pp.17, 31–33, 39, 41, 61). Residual: <code>E34.lemma3</code> and <code>E34.transfer</code> are still plain text in MD l.2342/2348 (EN p.64–65; ES p.66).
X-13 Spanish subscripts in EN math	RESOLVED	far, cross, clean and covered appear in both languages; <code>mis_c</code> is defined.
X-14 literal Lean excerpt	RESOLVED	EN p.39 and ES p.40: strict-implicit binder $\{v : V\}$ and indentation are preserved.
X-15 untranslated English words in ES	PARTIAL	baseline/build/target/log/gap were translated. Residual: "matching(s)" × 12 in prose (e.g. l.526, 578, 746, 1845, 1870, 1912, 2130–2137), "ni otro packing" (l.688), and "Cotas de edición y baseline" inside the ES Figure 3 (rendered p.26).
X-16 "§3.2"	RESOLVED	A.2 row now reads "Lemma 3.2 and C.1–C.2" / "Lema 3.2 y C.1–C.2".

New findings (v1.21)

ID	Severity	Location	Description
NEW-02	MINOR (bibliographic)	ES md l.2484; ES PDF p.71	The translation pass altered a cited title. The Haxell–Rödl reference [6] now reads «Integer and fractional empaquetamientos in dense graphs». v1.2 ES and v1.21 EN have the correct title «Integer and fractional packings in dense graphs». The author's response says bibliographic titles were not translated, so this is an accidental corruption of the reference list.
NEW-01	MINOR	EN §5.1 heading (p.13) and A.2 row (p.40); ES Lemma 3.2 heading (p.44)	Descriptive words are typeset as code identifiers (<code>Regularization</code> , <code>Nibble</code>). The headings are inconsistent between languages: EN "5.1. Regularization..." vs ES "5.1. Regularización..."; ES "Lema 3.2. Nibble..." vs EN "Bounded-rank nibble".
OBS-1	OBSERVATION	EN §7.2	"the bounded gain function from Model" lost its code markup, although it names the Lean namespace <code>Model</code> .
OBS-2	OBSERVATION	ES p.33	<code>#check</code> is rendered with letter-spacing; the text remains legible.
OBS-3	OBSERVATION	EN p.70 / ES p.71	The reference [5] note writes "PreferenceLabs"; the README writes "Preference Labs".

E6 verdict: PASS with findings. No statement, number or qualifier differs between EN and ES. One new MINOR bibliographic error (NEW-02). Two earlier findings are only partially resolved (X-12, X-15).

PDF generated by pandoc (HTML) + PyMuPDF Story from the Markdown source; not compiled from TeX. The TeX file alongside is pandoc's standalone conversion of the same Markdown.